expirement95g6g.m | exp105g.m | exp115g6g.m | exp125g6g.m

/MATLAB Drive/exp125g6g.m

```matlab
clc;
clear;
close all;

% SNR values (dB)
SNR = 0:5:30;

% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M = 1.2*(1-exp(-SNR/8));

disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
disp([SNR' NB_IoT' LTE_M'])

% Simulated Latency (ms)
NB_IoT_Latency = [180 160 145 130 120 110 100];
LTE_M_Latency = [90 80 70 60 50 45 40];

disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')
disp([SNR' NB_IoT_Latency' LTE_M_Latency'])

figure

% -------- Graph 1 --------
subplot(3,2,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on

% -------- Graph 2 --------
subplot(3,2,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on

% -------- Graph 3 --------
subplot(3,2,3)
plot(SNR,NB_IoT_Latency,'-o','LineWidth',2)
title('NB-IoT Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
```

Figure 1